

SPMM
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Child Psychiatry

Paper B

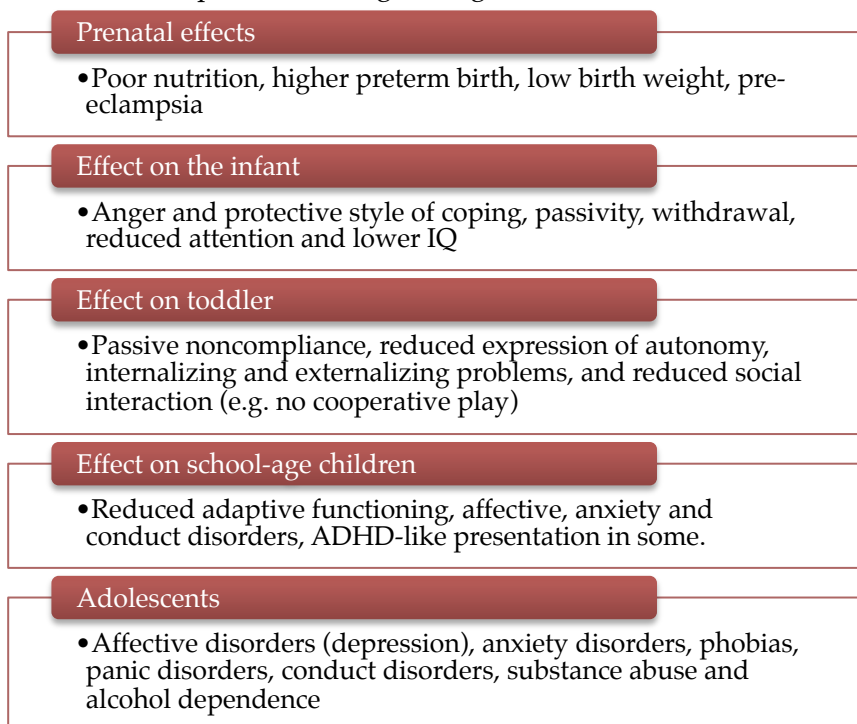
Syllabic content 10

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1. Adult mental illness and children

- ★ The consequences on the child of parental mental illness affect their infancy, toddlerhood, preschool age, school age and adolescence. The best-studied example of the influence of parental illness on child's mental health is that of maternal depression.
- ★ It is important to note that the effects of parental mental illness on children are mediated by complex factors.
- ★ Rutter has outlined a number of possible mediators of the effect of parental psychiatric disorder on a child. First, the direct pernicious impact of exposure to the parental disorder. Second, an indirect impact due to altered interpersonal behaviour and parenting capacity. Third, mediator variables, such as the social adversity, genetic or constitutional factors could play a part (Murray & Cooper, 1997).
- ★ Consequences of maternal depression on a growing child is summarised below:



Adapted from Canadian Paediatric Society, Maternal depression and child development. (2004). *Paediatrics & Child Health*, 9(8), 575–583; Murray, L & Cooper, PJ.

2. Negative life events during development

- ★ Childhood adversities are significantly associated with adult mental disorders as shown in numerous epidemiological studies.
- ★ The association between retrospectively reported childhood adversities and the first onset of a wide variety of mental disorders across the life course was studied in epidemiological surveys in 21 countries in the World Health Organization (WHO) World Mental Health (WMH) Survey Initiative (Kessler et al., 2010). Some notable findings from this large study are as follows:
 - Parental death is the most common childhood adversity (11.0–14.8%).
 - Other adversities included physical abuse (5.3–10.8%), family violence (4.2–7.8%) and parental mental illness (5.3–6.7%).
 - Multiple childhood adversities (mean = 2 or more in nearly 2/3rd) were common among respondents with any childhood adversities.
 - Adversities associated with maladaptive family functioning elevate the risk of adult psychiatric disorders (all major DSM-IV disorders) with an odds ratio of 1.3–2.4; the risk was lower and often insignificant for other types of childhood adversities. These results were strikingly consistent globally.

Child Abuse

- ★ Physical abuse involves the intentional injury of a child by someone. Physical abuse can be defined as any act that results in a nonaccidental physical injury, such as beating, punching, kicking, biting, burning, and poisoning.
- ★ Sexual abuse of children refers to sexual behaviour between a child and an adult or between two children when one of them is significantly older or uses coercion. The perpetrator and the victim may be of the same or opposite sex. The sexual abuse may be non contact or contact abuse and may include exhibitionism; nongenital or genital fondling; fellatio; cunnilingus; or vaginal or anal penetration.
- ★ Depression, PTSD, conduct disorders, somatisation and suicidal behaviour is seen in those with a h/o sexual abuse. Girls are more often the victims [4:1], 90% abusers are male. Avg age 9-11 years depending on the study.
- ★ Psychological abuse or emotional abuse occurs when an adult repeatedly conveys to a child that he or she is worthless, defective, flawed, unloved, or unwanted, flawed or endangered.
- ★ Neglect, the most prevalent form of child maltreatment, is the failure to provide adequate care and protection for children. Malicious or ignorant withholding of physical, emotional, and educational necessities can harm children. Neglect includes failure to feed children adequately and to protect them from danger.
- ★ Of the cases reported, approximately 60% are for neglect, 20% for physical abuse, 10% for sexual abuse, and 10% for other forms of maltreatment (e.g., psychological abuse, abandonment etc)
- ★ Child maltreatment is a non-specific risk factor for multiple forms of psychopathology. They also have increased rates of posttraumatic stress disorder, depression, self-destructive behaviour and borderline traits, sexually inappropriate behaviours, drug and alcohol problems, eating disorders, oppositional defiant disorder and conduct disorder.

Signs of possible physical abuse

- Unexplained injuries, bruises or burns, especially if recurrent
- Improbable excuses given for injuries
- Refusal to discuss injuries
- Untreated injuries or delay in reporting them
- Excessive physical punishment

Signs of possible physical neglect

- Constant hunger
- Poor personal hygiene
- Constant tiredness
- Poor state of clothing
- Frequent lateness or non-attendance at school
- Untreated medical problems

Signs of possible non-organic failure to thrive

- Significant lack of growth
- Weight loss and hair loss
- Poor skin or muscle tone and circulatory disorders

Signs of possible emotional abuse

- Low self-esteem, continual self-deprecation
- Sudden speech disorder
- Rocking, head-banging, or other neurotic behaviour
- Self-mutilation

Signs of possible sexual abuse**Behavioural**

- Lack of trust in adults or over-familiarity with adults
- Fear of a particular individual
- Social isolation, withdrawal, and introversion
- Sleep disturbances (nightmares, irrational fears, bed wetting, fear of sleeping alone, needing a nightlight)
- Running away from home
- Girls taking over the mothering role
- Unusual interest in genitals of adults, children, or animals
- Expressing affection in inappropriate ways
- Developmental regression
- Over-sexualised behaviour

Physical/medical

- Bruises, scratches, or other marks to the thighs or genital area
- Itch, soreness, discharge, unexplained bleeding from the rectum, vagina, or penis
- Pain on passing urine or recurrent urinary tract infection
- Recurrent vaginal infection
- Venereal disease
- Stained underwear
- Discomfort/difficulty walking or sitting
- Pregnancy, particularly when reluctance to name father
- Higher morning cortisol levels, than non abused children

Adapted from; Oxford Handbook of Psychiatry, 1st Edition

- ★ Factitious disorder/Munchausen's syndrome by proxy: when a person feigns or induces illness in a child or others in order to obtain medical attention. A form of child abuse. Can be hard to detect as the perpetrator may cover up or deny behaviour. Need to be aware when a young person repeatedly presents for medical attention.

Aetiological factors for abuse

Physical Abuse

- ★ Although child abuse occurs at all socio-economic levels, it is highly associated with poverty and psychosocial stress, especially financial stress.
- ★ Physical abuse most commonly begins around adolescence. Child maltreatment is strongly correlated with less parental education, underemployment, poor housing, welfare reliance, and single parenting.
- ★ Child abuse tends to occur in multiproblem families, that is, families characterized by domestic violence, social isolation, parental mental illness, and parental substance abuse, especially alcoholism.
- ★ Risk factors in the parent(s): young age, low IQ, criminal record, poor parenting skills, experience of abuse/neglect as a child and psychiatric problems.
- ★ Risk factors in the child include prematurity, congenital malformation, intellectual disability, chronic illness, difficult temperament (eg. crying a lot).
- ★ The risk of child abuse increases in families with many children.

Sexual Abuse

- ★ Socioeconomic status is unrelated to the incidence of child sexual abuse, but lower socioeconomic classes are more likely to come to attention.
- ★ There is an inverse relationship between self blame and powerlessness in sexually abused children.
- ★ Most common relationship is stepfather and stepdaughter
- ★ Social, cultural, physiological, and psychological factors all contribute to the breakdown of the incest taboo. Incestuous behaviour has been associated with alcohol abuse, overcrowding, increased physical proximity, and rural isolation that prevent adequate extra familial contacts.

3. Attention Deficit Hyperactivity Disorder

Clinical features of ADHD

- ★ The cardinal features of ADHD are excessive and impairing levels of hyperactivity, inattention, and impulsivity, and they are pervasive over time.
- ★ These features must be evident in more than one setting, cause serious impairment and be excessive in relation to the person's mental age and development, and must not be due to other causes such as anxiety, schizophrenia or autism.
- ★ Hyperactivity-impulsivity symptoms include: fidgeting, being 'always on the go', talking excessively; unable to play quietly; continually interrupting.
- ★ Inattention symptoms include: being easily distracted, being unable to sustain attention, difficulties completing tasks, difficulties organizing, avoiding tasks requiring mental effort; appearing not to listen, being forgetful, and losing things.
- ★ Hyperactivity is more impairing and more noticeable in pre-school children. In school inattention is more noticeable, and inattention and impulsivity is noticeable in both adolescent and adult populations, especially in social situations.
- ★ Connor's questionnaire is widely used to obtain information from schoolteachers on ADHD symptoms. There are also parent and adolescent versions
- ★ **Diagnostic criteria: The hyperkinetic disorder** is the ADHD equivalent in ICD-10. For ADHD/HKD, the diagnostic criteria are considered to be more 'relaxed' in DSM but stricter in ICD-10. According to DSM-IV criteria, to meet the diagnosis of ADHD, some symptoms must be present before the age of 7 years, although ADHD is not diagnosed in many children until they are older than 7 years when their behaviours cause problems in school and other places.
- ★ To confirm a diagnosis of ADHD, impairment from inattention and/or hyperactivity-impulsivity must be observable in at least 2 settings and interfere with developmentally appropriate functioning socially, academically, or in extracurricular activities and should persist for at least six months.
- ★ ADHD is not diagnosed when symptoms occur in a child, adolescent, or adult with a pervasive developmental disorder, schizophrenia, or another psychotic disorder.

DSM-5 AND ADHD

For ADHD the onset criterion has been changed from "symptoms that caused impairment were present before age 7 years" to "several inattentive or hyperactive-impulsive symptoms were present prior to age 12"

Subtypes have been replaced with presentation specifiers that map directly to the prior subtypes

A comorbid diagnosis with autism spectrum disorder is now allowed

The symptom threshold has been changed for adults with a cutoff for ADHD of five symptoms, instead of six required for younger persons, both for inattention and for hyperactivity and impulsivity.

Prevalence:

- ★ Using DSM-IV criteria, proposed general prevalence in school age children is about 5%. Using ICD-10 criteria, prevalence is approximately 1-2% (Shorter Oxford Textbook of Psychiatry, 6th Edition).
- ★ The prevalence in UK children, using DSM IV criteria is 3-4% (The British Child and Adolescent Mental Health Survey 1999: the prevalence of DSM-IV disorders)
- ★ ADHD is 3 times more prevalent in boys than in girls, and more common in areas of social deprivation and amongst children living in institutions

Aetiology:

- ★ **Genetics:** Siblings have 2-3 times increased risk; heritability of approximately 80%. Greater concordance in monozygotic compared with dizygotic twins [concordance 79% in monozygotic and 32% in dizygotic twins]. Genes 5, 6, and 11 implicated. This condition is associated with dopamine transporter gene [DAT1] and dopamine D4 receptor gene. SNAP-25 gene may also have a role.
- ★ **Neuroimaging:** Areas of brain affected include: prefrontal cortex, striatum and cerebellum. Studies using positron emission tomography (PET) have found lower cerebral blood flow and metabolic rates in the frontal lobe areas of children with ADHD than in controls. PET scans have also shown that adolescent females with the disorder have globally lower glucose metabolism than both normal control females and males with the disorder.
- ★ **Neurotransmitters:** DA and NA dysregulation in the prefrontal cortex is implicated. The most widely studied drugs in the treatment of ADHD, the stimulants, affect both dopamine and norepinephrine, leading to neurotransmitter hypotheses that include possible dysfunction in both the adrenergic and the dopaminergic systems. Serotonin may have a role in modulating dopamine transmission and the expression of ADHD.
- ★ Potentially important **environmental factors** include a) prenatal and perinatal obstetric complications b) low birth weight & prematurity c) prenatal exposure to alcohol, nicotine and benzodiazepines d) poor attachment and severe early deprivation e) institutional rearing
- ★ Idiosyncratic reactions to food, Other food additives, lead exposure to toxic levels are not supported by research evidence.
- ★ Previous research shows that ADHD occurs in head injury in 25% cases. Retrospective cohort study published in BMJ found that head injury before the age of 2 years does not seem to be causal in the development of ADHD. Medically attended injury before 2 years of age may be a 'marker' for subsequent diagnosis of attention deficit hyperactivity disorder.
- ★ The quality of relationships within the family and at school can be considered as protective or maintaining factors.
- ★ **Comorbidity:** 50-80% of children with ADHD have a comorbid disorder. 50% children may meet criteria of 2 comorbid conditions. Common comorbid problems in childhood include oppositional defiant disorder (40%), anxiety disorder (34%); conduct disorder (14%), tics (11%) and mood disorder (6%) (Young et al, 2011, BMC Psychiatry)

Outcome:

- ★ Approximately 15% of cases continue to meet diagnostic criteria for ADHD at the age of 25 years. (Young et al, 2011, BMC Psychiatry). A further 50% of individuals will suffer some impairment from residual symptoms.
- ★ Children with hyperkinetic disorders are 5 times greater risk for antisocial behaviour, substance abuse and other psychiatric disorders. 15-20% develop substance misuse problems
- ★ Many children initially diagnosed with ADHD, combined type, exhibit fewer impulsive-hyperactive symptoms as they get older and, by the time they are adults, will meet criteria for ADHD, inattentive type.
- ★ Poor prognosis depends on early stressful life experiences such as due poverty, overcrowding, expressed emotions and parental psychopathology. Prognosis is worse when the symptoms are severe, predominantly hyperactive-impulsive in nature, and associated with conduct, language or learning disorder.

Management

- ★ Treatment [long term, multimodal]: Before resorting to pharmacological treatments, efforts should be made to provide psychological and social interventions. If medication is being considered, the young person/family need to consent, and must be aware of the possible side effects of treatment.
- ★ General management principles are:
 - Educational/remedial interventions
 - Parent training programme for child management skills– based on social learning theory and behavioural interventions
 - Individual/family/group therapies
 - CBT methods, especially behavioural, are often effective
 - Social skills training
- ★ **MTA Study:** The Multimodal treatment study of children with ADHD was the largest, most rigorous randomised controlled trial to date, involving 579 children. Despite methodological issues raised by few authors, it is a trial that confirmed the effectiveness of medication management in children and adolescents. The trial also found that intensive behavioural therapy involving the child; family and teachers added little to well supervised medication management. However, psychological interventions are important for families who do not wish to use medication. Also, there is lack of evidence of efficacy over prolonged periods of treatment with medication.
- ★ **Stimulants** – release noradrenaline, dopamine and also serotonin, increasing extracellular dopamine and thus ‘inhibit’ impulses, helping persistence in motor and cognitive functions. They are rapidly absorbed following oral administration.
- ★ Methylphenidate has most rapid onset 1-3hr, and shortest half-life 2-3 hours so the effects wear off in 4-6 hours. Methylphenidate dose range – 5-60mg/day. Growth retardation may occur during acute treatment, but no significant retardation seen in long term. Note: Prescription of stimulants does not increase the rate of substance abuse by the patients.
- ★ Pimoline is no longer licensed in UK as it causes abnormal liver function tests, and in some cases liver failure following long term use.

- ★ **Atomoxetine** is a noradrenaline reuptake inhibitor – good evidence for use in hyperkinetic disorders. It increases noradrenaline in the synaptic cleft. It doesn't affect dopamine levels, and hence doesn't cause more tics. In December 2012, MHRA issued a note highlighting that Atomoxetine can cause clinically relevant increases in blood pressure or heart rate, or both, in a small proportion of patients and suggested monitoring heart rate and BP.
- ★ **Monitoring:** Height, weight, blood pressure, heart rate monitoring is recommended for methylphenidate (initially 3 monthly, then 6 monthly). Atomoxetine rarely causes liver damage; so routine monitoring of LFT is not warranted. The MHRA also advises that patients on Atomoxetine should be monitored for signs of depression, suicidal thoughts or suicidal behaviour. Monitoring of height and weight is also recommended, initially three monthly and then six monthly.

Stimulants (methylphenidate)		
1 st Line Medication	Specifically treat ADHD core symptoms [hyperactivity>inattention] Largest and most rapid effect on ADHD of any drug class, indirect sympathomimetic by increasing DA and AND release. Calms comorbid aggression and oppositional-defiant behaviour Except for pemoline, medically safer than most psychoactive drugs Results of given dose seen immediately; relatively easy titration Now available in XL preparation	Adverse effects: Appetite, weight loss Sleep disturbance (if taken late in day) Cramps (first few weeks) Headaches Mild BP and pulse increase Evening crash constricted affect and spontaneity, emotional blunting, social withdrawal, Depression Tics Hallucinations (skin crawling, visions) Mild growth slowing first 2 years Dose for behaviour may not be optimal for attention (normal dose: Initially 5-10 mg daily titrated up to a maximum of 60 mg/day in divided doses using weekly increments of 5-10 mg.)
Atomoxetine		
	Noradrenaline reuptake inhibitor Specifically treats ADHD core symptoms Nearly as effective as stimulants; may help stimulant failures Continuous duration of effect Little or no insomniac side effect Can be given any time of day –once daily No tic side effect (good choice with comorbid tics) May help comorbid depression	Adverse effects: Appetite, weight loss Gastrointestinal Sx (nausea, vomiting, diarrhea, constipation) Fatigue, dizziness Probable, mild growth slowing Allergic reactions Possibly longer time than stimulants to flush out if adverse effect Slower attainment of full effect than stimulants
Antidepressants		
Tricyclics	Treat both ADHD and comorbid depression and anxiety Helps some stimulant non-responders Third most effective drug class for ADHD (except for SSRIs, which are not very effective) Some patients/families who are prejudiced against stimulants will accept antidepressants May equal stimulant effectiveness for adults	Adverse effects: Sedation BP changes (down or up) Dizziness (especially on standing) Dry mouth Cardiac conduction block: TCAs require ECG monitoring in children

Alpha 2 agonists		
	Treat both hyperactivity-impulsiveness and comorbid tic disorder or comorbid aggression Helps some nonresponders to stimulants and antidepressants Good for those overaroused, possibly with comorbid anxiety	Adverse effects: Response delayed Sedation Hypotensive dizziness (especially postural) Dry mouth Rare hallucinations Hypertensive rebound if dose missed Not as helpful for attention as stimulants
Antipsychotics		
	May work when stimulant or atomoxetine does not, especially if stimulant makes worse Good for comorbid anxiety, aggression, tic disorder, or bipolar disorder	Adverse effects: Sedation Extrapyramidal side effects Endocrine effects Tardive dyskinesia Paradoxical agitation (akathisia) Weight gain Not specific, generally less effective than stimulants or atomoxetine Riskiest drug, last resort
Adapted from Lewis's Child & Adolescent Psychiatry: A comprehensive textbook 4 th edn		

4. Disorders of childhood conduct

Disorders of childhood conduct comprise of (a) Conduct Disorder and (b) Oppositional Defiant Disorder. Conduct disorder (CD) is characterized by a severe and persistent pattern of antisocial, aggressive or defiant behaviours that defy age-appropriate societal norms. Oppositional Defiant Disorder (ODD) also involves a persistent pattern of defiant behaviour. However, the behaviour in the latter does not defy age-appropriate societal norms to the same extent as in CD. According to ICD10, oppositional defiant disorder is a subtype of conduct disorder. DSM-5 excludes oppositional disorder if a conduct disorder is present.

Conduct Disorder

Diagnostic criteria:

- ★ Children with conduct disorder are likely to demonstrate behaviours in the following four categories
 - Physical aggression or threats of harm to people, cruelty to people and animals
 - Destruction of their own property or that of others
 - Theft or acts of deceit
 - Frequent and serious violation of age-appropriate rules (Like truancy or running away)
- ★ ICD-10 requires at least one behaviour to be present for at least six months. According to DSM-5 criteria, at least 3 out of a list of 15 behaviours should begin before the age of 13, for a period of 12 months. DSM-5 has added a *limited prosocial emotions specifier* to the diagnosis of conduct disorder for children who do not meet the full criteria but present with limited prosocial emotions, such as limited empathy and guilt. Other specifiers retained from DSM-IV are childhood onset type – symptoms present before age 10 and an adolescent onset type – symptoms develop after age 10.
- ★ CD is the cause of great suffering in both the individual and in society; it is one of the major risk factors for adult antisocial behaviour, posing a major burden on public resources. Conduct disorder occurs with greater frequency in the children of parents with antisocial personality disorder and alcohol dependence than in the general population. In the Isle of White study, CD was found to be the most common psychiatric disorder amongst 10-11 year olds.
- ★ A prevalence of 5-7% is noted in the UK (Oxford Handbook of Psychiatry, 3rd Edn.). The disorder is more common among boys than girls, and the male: female ratio is 4:1.

Aetiology:

- ★ Ontario Child Health Survey[1987] : Three most significant risk factors:
 - Family dysfunction
 - Parental mental illness
 - Low income
- ★ Rutter [1978] :
 - Low socioeconomic status, [Low family income]
 - Criminality of father,
 - Overcrowding,
 - Maternal neurosis,
 - Institutional care

- Chronic marital discord

★ **Biological risk factors:**

- *Genetic*: CD clusters in families. Heritable trait of being more susceptible to externalizing disorders in general (CD, ODD and and ADHD) (Shorter Oxford Textbook of Psychiatry, 6th Edition).
- *Temperament*: 'callous-unemotional'
- *Brain injury*
- *Neurochemical*: Low CSF serotonin, deficient serotonergic activity is seen in those with early onset and more aggressive behaviour. Autonomic under-arousal e.g., consistent lower mean resting heart rates, lower electro dermal activity etc. HR indices can predict later onset of aggression. Low salivary cortisol levels are also reported.
- *Low IQ*. Performance IQ may be higher than Verbal IQ
- *Neuroimaging*: Prefrontal brain regions may have reduced volumes in affected children.

★ **Psychosocial risk factors:**

- Maternal smoking during pregnancy
- Parental criminality and substance abuse
- Harsh and inconsistent parenting
- Lack of a warm parental relationship and cold/rejecting family relationships
- Domestic violence in the family and child abuse
- Large family size
- Low socio-economic status / Low family income and social disadvantage
- Early loss and deprivation
- School failure and poor school achievement
- Social isolation;
- Exposure to urban life does not increase the rate of conduct disorders, though some studies have found higher prevalence rates in urban locations

Outcome

★ Childhood conduct disorders further predict risk for numerous problems in adulthood that includes (Moffitt et al-2002)

- Criminality and antisocial personality disorder. Less than 50% of conduct disorder children have persistent and severe antisocial problems as adults (Zoccolillo et al 1992).
- Serious difficulties in education, work and finances
- Homelessness and abuse
- Drug and alcohol dependence
- Poor physical health including injuries, sexually transmitted infections, compromised immune function
- Variety of mental disorders and suicidal behaviour

Protective factors

- Female gender
- High IQ
- Resilient temperament
- Good parenting
- Warm relationship with a key adult
- Commitment to social values
- Increased economic equality.

Poor prognostic factors

- Early onset (before 10 yrs)
- Increased aggression at early age
- Aggression carried out in isolation and not in groups
- Low IQ
- Low socio-economic status
- Poor school achievement
- Attentional problems and hyperactivity as a child
- Poor parenting and family criminality.

Treatment:

- ★ As the causes and risk factors are multifactorial, treatment is multimodal. Explore specific support for academic and social skills. Psychological therapies form the mainstay of treatment for conduct problems. Parent Management Training based on the principles of social learning theory has been very successful in altering the course of conduct disorders. NICE recommends that group based parent training/education programmes should be the mainstay of treatment for children of 12 years and under with oppositional defiant disorder and conduct disorder.
- ★ *Cognitive behavioural therapy*: CBT for conduct problems in children and adolescents typically includes social skills training and anger management. The most common targets are aggressive behaviour, social interactions, self-evaluation and emotional dysregulation.
- ★ *Functional family therapy*: One of the best-known interventions for serious antisocial behaviour is functional family therapy. It is designed to be practicable and relatively inexpensive. The target age range is 11-18 years. Between eight and twelve 1-hour sessions are given in the family home to overcome attendance problems. For more intractable cases, 12-16 sessions are offered and it usually lasts for three months. There are four phases of treatment which includes Engagement, Motivation, Behavioural Change and Generalisation. The aim is first to keep the family in treatment and only then to move on to finding what precisely they want. The therapist must understand the parents' goals before specific techniques are taught. Functional family therapy addresses family processes, which need to be present, such as improving communication between parent and young person, reducing interparental inconsistency, tightening up on supervision and monitoring, and negotiating rules and the sanctions to be applied for breaking them. Functional family therapy has been shown to reduce reoffending rates by around 50 per cent.
- ★ *Multisystemic therapy*: It is one of the best-developed treatments of conduct disorder, which rests on nine treatment principles. The clinicians take on only four to six cases at a time and the team is available 24 hours a day. Treatment is usually given for three months and then stopped. In

Multisystemic therapy the young person's and family's needs are assessed in their own context at home and in their relationships with other systems such as school and peers. Following the assessment, proven methods of intervention is used to address difficulties and promote strengths. In this therapy, assessing and promoting the strengths in the young person and the system is very important. The therapist is responsible for ensuring appointments are kept and for making change happen—families cannot be blamed for failing to attend or 'not being ready' to change. Regular written feedback on progress towards goals from multiple sources is gathered by the therapist and acted upon and the parents and teenagers fill in weekly questionnaires on whether they have been receiving therapy as planned. There is close attention to 'quality control' by offering weekly supervisions and the supervisor checks adherence weekly. Several randomised controlled trial attest to effectiveness, with reoffending rates typically cut by half and time spent in psychiatric inpatient care further reduced.

★ *Other treatment options:*

- Treat any comorbidity
- Address any child protection concerns
- Anger Management Programmes

Oppositional Defiant Disorder

- ★ ODD is characterised by an enduring pattern of negative, hostile, disobedient and defiant behaviour, without serious violations of societal norms or the rights of others. Symptoms must be persistent and evident for at least 6 months. In oppositional defiant disorder, a child's temper outbursts, active refusal to comply with rules, tendency to blame others, spiteful and annoying behaviours exceed expectations for these behaviours for children of the same age.
- ★ Manifestations of the disorder are almost invariably present in the home, but they may not be present at school or with other adults or peers. In some cases, features of the disorder from the beginning of the disturbance are displayed outside the home; in other cases, the behaviour starts in the home, but is later displayed outside. Typically, symptoms of the disorder are most evident in interactions with adults or peers whom the child knows well.
- ★ Although ODD can begin as early as 3 years of age, it typically is noted by 8 years of age and usually not later than adolescence. Age of onset of ODD is generally earlier than that for CD. Prevalence 2-5% (Oxford Handbook of Psychiatry, 3rd Edn). The disorder seems more prevalent in boys than in girls before puberty, and the sex ratio appears to be equal after puberty. 25% cases show no symptoms in later life (Oxford Handbook of Psychiatry, 3rd Edn) but many progress to conduct disorder
- ★ Aetiology includes temperamental factors like sick or traumatised child and power struggle between parents & child.
- ★ Longitudinal studies suggest that ADHD in early life is a predictor of oppositional defiant disorder and conduct disorder later in life.
- ★ Chronic oppositional defiant disorder almost always interferes with interpersonal relationships and school performance. Secondary to these difficulties can be low self-esteem, poor frustration tolerance, depressed mood, and temper outbursts. Adolescents may abuse alcohol and illegal substances. The disturbance may evolve into a conduct disorder or a mood disorder

- ★ Poor prognosis is associated with early onset of symptoms, longer duration of symptoms, co-morbid anxiety, impulse control & substance misuse disorders and development of conduct disorder.
- ★ **Management:** The principles are the same for both ODD and CD. The primary treatment of ODD is family intervention using both direct training of the parents in child management skills and careful assessment of family interactions. With the child, use methods such as collaborative problem solving skills. Behaviour therapists emphasize teaching parents how to alter their behaviour to discourage the child's oppositional behaviour and encourage appropriate behaviour. Behaviour therapy focuses on selectively reinforcing and praising appropriate behaviour and ignoring or not reinforcing undesired behaviour.

5. Early onset mood and psychotic disorders

Depressive disorder

- ★ The same diagnostic criteria for a major depressive episode apply in both adult and youth populations, but the way that young people can present can vary (see 'clinical features' section below). Core symptoms of low or irritable mood, fatigue or anhedonia, along with at least five other symptoms, such as social withdrawal, worthlessness, guilt, suicidal thoughts or behaviour, sleep increase or decrease, decreased motivation and/or concentration, and increased or decreased appetite.
- ★ The prevalence of depression amongst young people is increasing. Pre-puberty: about 1%, no sex difference; Post-puberty: about 3%, commoner in females.
- ★ 50% of young people with depression remain clinically depressed after 12 months. Most young people with major depression will recover from the episode within 2 years. Adolescent depression however shows significant continuity [eg 30% will have a recurrence within 5 years] into adulthood.
- ★ The single most important distinction between depression as an illness and the normal ups and downs of childhood and adolescence is that depression is associated with functional impairment, mediated through the intensity, duration, and lack of responsiveness of depressed mood and associated symptoms.
- ★ The full, pure clinical syndrome of depression is very uncommon before puberty. In early adolescence, cognitive changes such as formal operational thought allow hopelessness to be experienced and a clinical picture akin to adult depression can be seen. Depression should only be diagnosed when there is impairment of social role functioning, or with symptoms leading to significant suffering, or psychopathy eg suicidality.
- ★ It is important to rule out organic pathology and ask about possible substance misuse/abuse .

Clinical features:

- ★ *Young children:* Poor feeding, failure to thrive, tantrums, irritability, separation anxiety, hyperactivity, regressed behaviour such as enuresis, soiling etc.
- ★ *Older children:* Somatisation like pain in head, abdomen, chest and/or Hypochondriacal ideas, school refusal, Poor academic achievement at school, decline in school work, sleep disturbance, antisocial behaviour

- ★ *Adolescents:* Low mood may not be fixed and is more influenced by environment. Anhedonia and social withdrawal are powerful indicators of the presence of depression. Other features may include Low self-esteem, biological symptoms, suicidal acts, behavioural problems, and substance abuse.
- ★ Serious depressive episodes are rare before puberty, increasing in mid and late adolescence to adult levels. Clinical features are essentially the same as in adulthood but impairment may not be as salient, as in adults.
- ★ Dysthymic disorder is a chronic condition with fewer symptoms than major depression, but lasts a minimum of one year.
- ★ *Risk factors:* 95% of major depression in young people is seen in children with longstanding psychosocial difficulties. Other risk factors include
 - Family history of depression
 - Early loss of a parent
 - Parental separation, divorce and marital conflict
 - Stressful life events
 - History of abuse (physical, emotional and sexual)
- ★ *Maintaining factors*
 - Persistence of subthreshold symptoms
 - Scarring: First episode of depression sensitizes people to further episodes
 - Personality, temperament, cognitive abilities
 - Persisting adversity
 - Comorbidity
- ★ *Comorbidity:* Comorbidity is the rule (50-80%) rather than the exception in depressed children and adolescents. Anxiety [50 – 80%] is frequently a precursor of mood disorder and may also occur simultaneously with depression. Conduct disorder – 25% , OCD – 15%, eating disorder – 5%
- ★ ADHD and depression are also often co morbid and the two disorders may be co transmitted in families. Alcohol, drug, and tobacco abuse and conduct disorder are associated with depression

Treatment

- ★ Establish therapeutic alliance and provide education to the child and family. A multidisciplinary approach should be adopted.
- ★ Mild depression: 2 weeks of watchful waiting. After 4 weeks – supportive therapy, self help or group CBT. With self help consider, advice on exercise, sleep hygiene and anxiety management to the young person. Moderate to severe depression – CAMHS review, +/- 3 months of individual CBT, IPT, or shorter term family therapy. Consider an alternate psychotherapy if first mode doesn't help. CBT can reduce the duration of illness as compared to the other psychological treatments. NICE guidance for moderate to severe depression recommends psychotherapy before considering pharmacotherapy.
- ★ Combination treatment should be considered in all cases (i.e. psychotherapy + medication).
- ★ Several studies have demonstrated efficacy with selective serotonin reuptake inhibitors (SSRI) antidepressants, especially using fluoxetine. NICE guideline recommends fluoxetine as first line and sertraline or citalopram as second line. Fluoxetine is the best-studied antidepressant with the strongest

efficacy data, and consequently is the only antidepressant to receive FDA and MHRA approval and has been allowed by CSM for use for the treatment of depression in children and adolescents. Monitor for emergence of agitation, irritability, unusual changes in behaviour and emergence of suicidality at initiation and when dosages are changed. Paroxetine – evidence shows that Paroxetine has little impact to treatment, symptom levels, functional status or clinical improvement. Paroxetine is also more likely to bring serious adverse effects [and discontinuation] and increased suicidal behaviour.

- ★ TCAs are not supported by trials and have been associated with cardiac toxicity.
- ★ Psychotherapy is considered as a reasonable initial treatment in mild to moderate depression and this may be individual, group, or family therapy. CBT is one of the most researched techniques, but IPT, behaviour therapy, psychodynamic therapy, and supportive therapy are used.
- ★ *Treatment of Adolescents with Depression Study (TADS)* : TADS is the only published study to compare CBT, fluoxetine, their combination, and placebo. In this large [n=439], well-powered investigation, CBT (43% significantly improved) was not superior to placebo (35%), whereas both combination (71%) including fluoxetine, and fluoxetine alone (61%), were markedly superior to both CBT and to placebo. The combination treatment (CBT + fluoxetine) showed a faster recovery than any of the other treatments, although fluoxetine alone had a favourable outcomes with respect to the Clinical Global Impression Improvement (CGI-I) and baseline-adjusted endpoints and in more severely depressed patients. Combined treatment was superior to fluoxetine alone with regard to remission (37 vs. 20%).
- ★ A possible explanation for the relatively weak performance of CBT in this study could be the nature of TADS treatment package, which attempted to deliver a large number of approaches such as problem solving, behaviour activation, cognitive restructuring, emotion regulation, relaxation training etc in a brief period.
- ★ ECT not recommended for 5-11 year olds. In older persons, only in severe depression, life threatening symptoms, or severe intractable symptoms not responding to other treatment.

Suicide and Deliberate Self Harm:

- ★ Suicide is the third leading cause of death for adolescents, following accidents and homicides (Hawton 1986). 12% of adolescent deaths are due to suicide. Male suicides outnumber females at all ages and in all cultures.
- ★ Suicidal ideation is very common in adolescents – 14% boys, 25% girls. Suicide attempts and deliberate self harm (DSH) are more common in females, completed suicides are more common in males.
- ★ Few adolescents who make suicide attempts actually have psychiatric disorder and serious suicidal intent is low. In contrast, almost all children and teenagers who commit suicide suffer from a psychiatric disorder at the time of death – mood disorder, alcohol/substance abuse, anxiety disorders, or conduct disorders.
- ★ Self-poisoning is the commonest form of parasuicide-the usual agents, being analgesics and psychotropic agents. Others include attempted hanging, jumping from heights, car exhaust fumes and shooting being most frequent. Hanging is the most common method used by boys who completed

suicide. Overdose or jumping from heights is the most common method used by girls who completed suicide.

- ★ 10% of adolescents who attempt suicide repeat within a year. 40% suicides will have made a previous suicide attempt.

Comparison of suicide and non-fatal deliberate self-harm

	Completed suicide	Non fatal DSH
Incidence	Declining until recently	Rising
Sex	Males	Female
Age	Older (rare in children)	Younger (rare in children)
Social class	Upper and lower	Lower
Childhood	Death of a parent	Broken home
Precipitants	Guilt, hopelessness	Situational crisis
Family History	2-4 times likely to have a 1 st degree relative who committed suicide	
Setting	Often premeditated and alone	Impulsive and often others present

Bipolar disorder

- ★ The symptoms of mania commonly reported in children include: increased energy, distractibility, pressured speech, irritability, grandiosity, racing thoughts, decreased need for sleep, euphoria/elation, flight of ideas and poor judgement.
- ★ Children experiencing a manic episode typically present with atypical or mixed features characterized by irritability, labile mood and behavioural problems. Mood lability is a common and impairing presentation in youth, which may not always be a sign of mood disorder.
- ★ Elated/euphoric mood and irritability are the two symptoms with the most variability in rates across studies. However, though there was also significant heterogeneity in the rates of decreased need for sleep, racing thoughts, poor judgment, pressured speech, and distractibility.
- ★ **Epidemiology:** Prevalence in adolescents is approximately 1% and as high as 6% when including soft subsyndromal symptoms. M>F in childhood cases; M=F in adolescents. Retrospective studies in adults with Bipolar Affective Disorder have consistently reported that up to 60% had the onset of their mood symptoms before the age of 20 years.
- ★ **Co-morbidity:** Approximately 70% have ADHD, 40% ODD, 30% anxiety disorder, 40% have substance misuse problems, 8% Tourette's syndrome and 3% bulimia nervosa.
- ★ **Outcome:** Early onset BPD has a poor outcome with 50% showing long-term decline in function. The course is often chronic and less responsive to treatment, with atypical and rapid-cycling features especially difficult to treat.
- ★ Some studies suggest that approximately 20% of adolescents who have an episode of major depression experience a subsequent manic episode by adulthood. In depressed children and adolescents, the following features predict subsequent development of mania
 - Rapid onset depressive episode with psychomotor features

- Depressive episode with psychotic features
- Family history of mood disorder, especially mania
- History of hypomania or mania following treatment with antidepressant medications
- ★ Suicide risk is high with rates of completed suicide approximately 10%.
- ★ **Treatment:** NICE guidelines on bipolar disorder in children & adolescents recommend using adult medication guidelines but in **lower doses**. Atypical antipsychotics (Olanzapine, Risperidone) are preferred first line for acute mania followed by valproate/lithium. For maintenance the same choice of medication that controlled manic symptoms is preferred.
- ★ Polycystic ovaries and associated infertility are particular concerns when valproate is used for adolescent girls and NICE recommends avoiding its use in girls and young women.
- ★ As children have a higher renal filtration rate, and higher proportion of body water, higher doses of lithium (mg/kg of body weight) may be needed [to achieve a level of 0.4-1.0 mEq/L]. In younger children, neurological side effects such as tremors, drowsiness, ataxia and confusion may be seen more often than in adults.

Childhood onset schizophrenia:

- ★ Childhood-onset schizophrenia (COS) is a rare and severe form of schizophrenia characterized by an onset of psychotic symptoms by age 12 years, believed to represent a subgroup of affected individuals with an increased heritable etiology.
- ★ The term early-onset schizophrenia is used for onset before age 18. Very early-onset schizophrenia (VEOS) refers to onset before 13 yrs

Epidemiology:

- ★ Schizophrenia in prepubertal children is exceedingly rare; In adolescents, the prevalence of schizophrenia is estimated to be 50 times that in younger children, with probable rates of 1 to 2 per 1,000. Boys seem to have a slight preponderance among children diagnosed with schizophrenia and are often identified at a younger age than girls.
- ★ In very early onset schizophrenia, onset before age 13, the sex ratio is approximately 2:1 with males predominating.
- ★ All of the symptoms included in adult-onset schizophrenia may be manifest in children with the disorder. Childhood onset schizophrenia is characterised by more negative symptoms, disorganised behaviour, greater disorganisation both of thought and sense of self and fewer systematized or persecutory delusions. It also runs a more chronic course with less chance of a full recovery. The onset is frequently insidious; after first exhibiting inappropriate affects of unusual behaviour, a child may take months or years to meet all of the diagnostic criteria for schizophrenia.
- ★ Premorbid course: EOS cases have **increased delays in language, reading, bladder control, and social functioning** than adult-onset cases. Reports have documented marked neuropsychological deficits in attention, working memory, and premorbid IQ among children who develop schizophrenia and its spectrum disorders. These premorbid manifestations also indicate a genetic/developmental component to schizophrenia

Clinical features

- ★ Children who eventually meet the criteria often are socially rejected and clingy and have limited social skills. They may have histories of delayed motor and verbal milestones and do poorly in school, despite normal intelligence. They often have an insidious onset. Delusions and hallucinations are prominent especially in those who present to services. Visual hallucinations are experienced by a significant number of children with schizophrenia and are often frightening. Delusions are present in more than one half of children with schizophrenia; the delusions take various forms, including persecutory, grandiose, and religious. Delusions increase in frequency with increased age. Blunted or inappropriate affects appear almost universally in children with schizophrenia. Children with schizophrenia may giggle inappropriately or cry without being able to explain why. Formal thought disorders, including loosening of associations and thought blocking, are common features among children with schizophrenia. Illogical thinking and poverty of thought are also often present. These clinical symptoms are associated with poor premorbid function with developmental delays.
- ★ It is important to rule out organic conditions (such as TLE, Thyroid disease, brain tumour, Wilson's disease).

Aetiology:

- ★ Genetic factors: Schizophrenia is known to be up to eight times more prevalent among first-degree relatives of those with schizophrenia than in the general population. Heritability estimates are as high as 82%. 5-10% risk amongst first degree relatives of people with schizophrenia compared with 0.2-0.6% amongst first degree relatives of controls. Adoption studies of patients with adult-onset schizophrenia have shown that schizophrenia occurs in the biological relatives, not the adoptive relatives. Genetic evidence is supported by higher concordance rates for schizophrenia in monozygotic twins than in dizygotic twins; cytogenetic abnormalities are more common in early onset cases than in Adult onset cases.
- ★ Brain imaging: enlarged lateral ventricles compared with AOS; Recent studies have documented grey matter loss in the brains of children with COS that started in the parietal region and proceeded frontally to dorsolateral prefrontal and temporal cortices, including superior temporal gyri.
- ★ Obstetric complications similarly implicated as in AOS. (maternal infections like rubella, obstetric complications eg hypoxia)
- ★ High expressed emotion, characterized by overly critical responses in families, has been shown to be correlated with increased relapse rates among patients with schizophrenia
- ★ Early attentional deficits, deficits in social functioning, deficits in organisational ability and a lower intellectual ability at age 16 and 17 predict later development of schizophrenia in at-risk individuals. (The Oxford Handbook of Child and Adolescent Psychiatry-219)
- ★ **Course and outcome:** Childhood-onset schizophrenia appears to be a more malignant type of schizophrenia, which presents a greater challenge to treat with pharmacology and psychosocial interventions. It seems to respond less to medication than schizophrenia with adult onset or adolescent onset, and the prognosis may be poorer.

- ★ Important predictors of the course and outcome of early-onset schizophrenia include the child's level of functioning before the onset of schizophrenia, the age of onset, IQ, duration of first episode, duration of untreated psychosis, presence of negative symptoms, response to pharmacological interventions, how much functioning the child regained after the first episode, and the amount of support available from the family.
- ★ Risk of premature death is higher than in adult form of the disorder. 8.5% in a study of 106 cases. The risk of suicide or accidental death as a result of psychotic symptoms appears to be about 5%. (Adult suicide rate for schizophrenia is 10%)

Treatment:

- ★ Algorithms for treating psychosis in young people are the same as those for adult patients. Metabolic side effects are more common in this age group and more intensive monitoring is required.
- ★ Atypical antipsychotics favoured over typical ones. Although first generation drugs such as haloperidol are effective, young people are more prone to EPSEs than adults. Drug induced akathisia is less common in children, but acute dystonia is more common.
- ★ Randomised controlled trials of antipsychotics in early onset psychosis suggest that olanzapine and risperidone are effective, with advantages over Haloperidol. Atypicals are commonly used as first-line agents in EOS.
- ★ Clozapine seems to be effective in treatment resistant psychosis in adolescents although this population may be more prone to neutropenia and seizures than adults.
- ★ Avoid depot and sedating antipsychotics
- ★ Psychosocial treatments should include family work and focus on psychoeducation, social skills, and problem-solving strategies and CBT methods.

6. Anxiety Disorders

- ★ Fear and anxiety represent normal reactions to danger. One critical differentiation between normal fears or anxiety and an anxiety disorder derives from the so-called impairment criterion: to receive an anxiety disorder diagnosis, there must be evidence of significant impairment or interference in the child's everyday functioning. Anxiety disorders adversely impact self-esteem, social relationships, and academic performance.
- ★ Behavioural avoidance is a primary area of impairment that might lead children to avoid many typical experiences enjoyed by peers. Anxiety or fear is also considered abnormal when the level of distress evoked by danger is considered extreme, relative to a child's peers

Prevalence

- ★ Prevalence rates seem to vary. Overall rates range from 5-15%, with 8% requiring clinical treatment (Oxford Textbook of Psychiatry 3rd edn.) Male: female ratio is equal in childhood; Prevalence increase in girls after adolescence when the female: male ratio is around 2:1. Prevalence rates (Oxford Handbook of Psychiatry, 3rd edn):
 - Separation anxiety disorder-3.5% in children and 0.8% in adolescents
 - Generalized anxiety disorder- approximately 4% of adolescents.
 - Simple Phobia- 10% in some studies; twice as common in females.
 - Social phobia-1% in children and between 5-15% in adolescents
 - Panic disorder-3-6%, peak onset 15-19 years.
- ★ Age of onset and presentation of symptoms vary for each disorder. Anxiety symptoms vary with age. Tearfulness and clinging in preschool children. Somatic complaints and hypochondriacal fretting are common in middle childhood (5-12 years) as are irritability and aggressive behaviour. Separation anxiety disorder and specific phobia usually have onset in early childhood. Generalized anxiety disorder occurs across all age groups, while obsessive-compulsive disorder, social phobia, and panic disorder tend to occur in later childhood and adolescence.
- ★ Co-morbidity: Children with SAD are more likely to have co morbid specific phobia than children with GAD or social phobia. In contrast, children with GAD and social phobia are more likely to have comorbid mood disorders as compared to children with SAD. Up to 90% of young people with GAD have a comorbid diagnosis- anxiety disorders, CD and substance abuse are most common. Depression is 8.2 times as likely in children with anxiety disorders than in children without anxiety disorders. Specifically, there is a significant link between GAD and depression that persists into adulthood.

Clinical presentations

Separation anxiety disorder

- ★ Separation anxiety disorder (SAD) is defined as developmentally inappropriate and excessive anxiety concerning separation from home or from those to whom the individual is attached. This anxiety will interfere with normal age-appropriate functioning. The essential clinical feature of separation anxiety is excessive worry about losing or being permanently separated from a major attachment figure.
- ★ *Symptoms*: anxiety about actual or anticipated separation from or danger to attachment figure; sleep disturbances and nightmares - Persistent reluctance or refusal to go to sleep without being near or next

to a major attachment figure, repeated nightmares about separation; Somatisation - Repeated occurrence of physical symptoms that includes nausea, vomiting, headache, stomach aches etc. especially on occasions that involves separation from a major attachment figure; School refusal (especially in adolescents) - Persistent reluctance or refusal to go to school because of fear of separation. School refusal and excessive somatic complaints in the context of actual or anticipated separations are the most common reasons for parents and children to seek treatment for SAD

- ★ The disturbance must last at least 4 weeks and cause clinically significant impairment in social, academic, and occupational domains. The onset must occur before 18 years of age, and it is generally not expected to occur below 6 years of age. If the onset occurs before the child is 6 years old, it is labeled as early onset separation anxiety disorder.

Attachment disorders:

- ★ *Attachment*: the emotional bond between child and caregiver. Develops through appropriate parental responses to child's behaviour. It allows the child to understand their 'inner world' and it is the foundation for safe separation and development of autonomy (Oxford Handbook of Psychiatry, 3rd edn).
- ★ Children exhibit considerable variation in their attachment relationships to their mothers. Different patterns of attachment have been described:
 - Secure (about 60%) – child uses carer as secure base but able to explore freely and go back to caregiver for comfort if necessary, carer sensitive to child's cues)
 - Insecure, avoidant/anxious type (about 15%) – appears interested in caregiver, minimal distress at separation, sometimes ignores/avoids caregiver
 - Insecure, ambivalent/resistant type (about 10%) – high levels of distress upon separation, resist comforting and can take a while to settle. Thought to be due to inconsistent care giving.
 - Disorganized/disorientated (about 15%) – child displays contradictory behaviour patterns. Thought to arise from either the child experiencing the caregiver as frightening or the caregiver being frightened themselves.
- ★ Mary Ainsworth created a way of studying the quality of the attachment of a child to their mother/caregiver- she developed the Strange Situation procedure (1978).
- ★ *Romanian Adoptees Study* – Data on severe attachment disorders have been obtained from follow up of children taken from severely deprived institutions in Romania and adopted by families in Canada or UK. 20% had severe disturbances at the age of 6. Duration of exposure to deprivation was strongly associated with severe disinhibited behaviour.

A. Reactive attachment disorder:

- ★ This disorder, occurring in infants and young children is characterized by persistent abnormalities in the child's pattern of social relationships, which are associated with emotional disturbance and reactive to changes in environmental circumstances.
- ★ It is diagnosed before the age of 5 years, and may manifest as either inhibited or disinhibited subtypes.
- ★ It is more common in poverty-stricken and socially disrupted environments.

- ★ Causes: It occurs as a direct result of severe parental neglect, abuse and serious mishandling. Other contributory factors include young, isolated, inexperienced, and/or depressed caretaker.
- ★ Features: Fearfulness and hypervigilance that do not respond to comforting are characteristic. Poor social interaction with peers is typical; aggression towards self & others is also common; growth failure occurs in some cases.

B. Disinhibited attachment disorder:

- ★ Disinhibited attachment disorder of childhood is characterised by a particular pattern of abnormal social functioning that arises during the first 5 years of life.
- ★ In the early stages, it is usually manifest by clinging and diffuse non-selectively focused attachment behaviour.
- ★ By the age of 4 years, diffuse attachment remains but clinging tends to be replaced by attention seeking and indiscriminately friendly behaviour.
- ★ During middle and later childhood this behaviour often persists and depending on circumstances, there may also be associated emotional or behavioural disturbance.

Sibling rivalry disorder

- ★ Onset is within 6 months of the birth of an immediately younger sibling. Duration of the disorder is at least 4 weeks. Emotional disturbance that is abnormal in degree and/or persistence
- ★ Emotional disturbance is shown by anxiety, regression, tantrums, and dysphoria; sleep difficulties, oppositional behaviour, or attention-seeking behaviour with one or both parents (two or more of these must be present). Behaviour can range from strong reluctance to share and a lack of positive regard to, in severe cases, overt hostility, physical trauma and/or maliciousness towards, and undermining of, the sibling (ICD-10).

School refusal

- ★ Refusal to go to school or stay in school, even when under pressure from parents and school authorities. This is not a psychiatric disorder per se - it can have many causes.
- ★ School refusal is considered as a problem rather than a diagnosis. It is a condition that can co-occur with anxiety and depressive disorders. Significant difficulty attending school, resulting in prolonged absence and/or severe emotional upset in children, characterizes school refusal.
- ★ These children often display excessive fearfulness, temper outbursts, or complaints of feeling ill when faced with the prospect of going to school. The complaint is more of physical symptoms such as headaches, abdominal pain, nausea, palpitations etc.
- ★ The nature of the anxiety associated with school refusal behaviour is likely to change with age, as is the nature of the precipitating events. For example, fear of separation is more common in younger school refusers, while in older children social-evaluative fears, such as fears of teachers or peers, are more commonly reported.
- ★ Incidence in children and adolescents reported as 1-5%. Prevalence uncertain. Sex distribution-equal
- ★ Three main incidence peaks by age:
 - 5-7 years: School entry. Possible separation anxiety
 - Age 11: Commonest age of presentation. May be triggered by transition to secondary school.

- Age 14 years and older: Possibly partly due to first presentations of depressive or anxiety features/disorders. Also need to consider bullying, pressure of exams or any specific stressors in individual cases.

School Refusers	Truants
1. Presence of emotional symptoms	Presence of antisocial symptoms
2. Family history of neurosis	Family history of antisocial behaviour
3. Over-protective parenting	Inconsistent discipline
4. Satisfactory academic achievement	Poor academic achievement
5. Small family or the youngest member	Large family size
6. Parents aware of child's absence	Child is neither at home, nor at school
7. No gender difference	Male predominance

- ★ In terms of outcome, 70% will successfully reintegrate into school, but a significant minority will not. Social anxiety and communication problems may persist.

Selective mutism

- ★ Characterized by a persistent failure to speak in specific settings, such as school (where there is an expectation to speak), despite full use of language at home or with family. Normally begins between ages of 3-5 years, after normal speech has been acquired.
- ★ Rates of associated psychiatric disorder, especially social phobia, high. A child with selective mutism may remain completely silent or near silent, in some cases whispering instead of speaking out loud.
- ★ Rare- affecting about 3-8/10,000 in the UK. Prevalence varies with age, with younger children at increased risk for the disorder. Persistent mutism affects less than 1 per 1000 children.
- ★ Slightly more common in girls.
- ★ Some cases will last for months to years. No evidence that any particular treatment is generally effective. A behavioural approach with positive reinforcement techniques aimed at increasing the frequency of talking and decreasing the frequency of non-communication will be helpful. It is important to ascertain what communication is like at home.

ANXIETY DISORDERS IN CHILDREN- KEY CHARACTERISTICS

Separation anxiety disorder (SAD)	Excessive anxiety concerning separation from loved one Possible risk factor for development of panic disorder or agoraphobia in adulthood
Social phobia (social anxiety disorder)	Persistent fear of social or evaluative situations Behavioural inhibition may be a temperamental predictor of social phobia in childhood or adulthood
Generalized anxiety disorder (GAD)	Excessive and uncontrollable worry about multiple issues At least one somatic complaint Close genetic link with depression
Specific phobia	Extreme fear of a specific situation or object Five types of phobias, some corresponding to evolutionary dangers (snakes, blood)
Panic disorder with or without agoraphobia	Unexpected panic attacks accompanied by worry about future attacks Agoraphobia is diagnosed if individual avoids places in which escape would be difficult or embarrassing Panic attacks can be caused by medical conditions (hyperthyroidism, cardiac abnormalities)

(Adapted from: Lewis's Child and Adolescent Psychiatry: A Comprehensive Textbook, 4th Edition)

Treatment

- ★ *Cognitive Behavioural Therapy (CBT)* has proven to be effective at treating anxiety disorders in children and adolescents. Most CBT interventions were initially developed for adult anxiety and phobic disorders. The techniques for child and adolescent populations are conceptually and structurally similar but modified according to the developmental level.
- ★ CBT technique focuses on relaxation training and cognitive restructuring. Individual CBT plus the family component was shown to effectively reduce anxiety symptoms and may have some added benefits, particularly for female children. Group CBT interventions have been shown to be effective for the treatment of socially phobic children and adolescents.
- ★ *Psychoeducation* and *parent training* are also useful.
- ★ *Psychodynamic therapies* include group, family, and individual/play techniques
- ★ *Pharmacological interventions*: SSRIs are first-line agents. Overall, efficacy and safety data support the use of selective serotonin reuptake inhibitors (SSRIs) to treat childhood social phobia, separation anxiety disorder and generalised anxiety disorder. Recent studies have demonstrated the efficacy of SSRIs for specific anxiety disorders. Benzodiazepines are not indicated as a routine treatment in children due to their potential adverse effects. There is also little or no evidence for the efficacy of other anxiolytics, such as buspirone or beta-blockers.

7. Other anxiety-related disorders

Post traumatic stress disorder:

- ★ When PTSD as a diagnosis was first formulated in 1980, it was initially believed that this would not be relevant to children! But it is now known that PTSD occurs frequently in children and adolescents with up to 6 percent of young people meeting criteria for this diagnosis at some point. The British National Survey of Mental Health reported that 0.4% of children aged 11-15 was diagnosed with PTSD, girls showing twice the rate for boys [Nice Guidelines].
- ★ Most common traumatic exposures for children and adolescents include physical or sexual abuse [can be chronic stress]; domestic, school or community violence; being kidnapped; terrorist attacks; motor vehicle or household accidents; or disasters, such as floods, hurricanes, tornadoes, fires, explosions etc. The child's response must involve intense fear, terror, helplessness, horror, or disorganized or agitated behaviour
- ★ *Classification of childhood trauma*
 - Type 1: Classic single, acute, traumatic event; the most common form in children. Symptoms: Full, detailed memories, 'omens' or 'cognitive reappraisal', misperceptions.
 - Type 2: Follows longstanding or repeated exposure to extreme external events, (e.g. chronic abuse). Symptoms: Denial and psychic numbing, self-hypnosis, depersonalisation, and dissociation and rage, self-harm, and/or extreme passivity

DSM-5 AND PTSD IN CHILDREN

Diagnostic threshold is lowered for children. In addition, a separate PTSD criterion has been added for children less than age 6.

Treatment

- ★ *Trauma-Focused Cognitive-Behaviour Therapy*: Several randomized clinical trials have provided evidence for the efficacy of trauma-focused cognitive-behaviour therapy (CBT) in the treatment of PTSD in children and adolescents. This treatment is generally administered over 8 to 12 treatment sessions, including a number of components. Group treatments are preferred for large number of people, with separate groups for girls and boys.
- ★ *Crisis intervention/psychological debriefing* has been adapted for groups of children following trauma. Role of crisis intervention with adults is being called into question. This involves structured sessions with group leaders discussing the trauma with a group of children to share feelings and knowledge and process it. Single session debriefing for children is not supported by evidence.
- ★ NICE guidelines advise that *medications* should not be routinely prescribed for children and young people with PTSD. Serotonin reuptake inhibitors (SSRIs) are frequently used in children with PTSD in the absence of evidence demonstrating efficacy. Citalopram from 20 mg to 40 mg has been reported to be helpful in the management of PTSD in children and adolescents according to results of an open trial over an 8-week period.
- ★ *EMDR* – no controlled trials – one unpublished trial with inconclusive results.
- ★ Families should be involved, where appropriate.

- ★ No good evidence for efficacy of other treatments, including play therapy, art therapy, or family therapy.

Obsessive compulsive disorder

OCD in children:

- ★ Obsessive-compulsive disorder (OCD) is characterized by the presence of recurrent intrusive thoughts (obsessions) associated with anxiety or tension and/or repetitive purposeful mental or physical actions (compulsions) aimed at reducing fears and tensions caused by obsessions. It has become increasingly evident that the majority of cases of OCD begin in childhood or adolescence.
- ★ The clinical presentation of OCD in childhood and adolescence is similar to that in adults and the only alteration for children is that they do not necessarily demonstrate awareness that their thoughts or behaviours are unreasonable. Childhood OCD is characterised by **secrecy** i.e. – they understand that their OCD behaviour is unusual, and so try to hide it. 40% children with OCD do not have obsessive thoughts.
- ★ Rituals and habits are seen in 2/3rd of children but they are less frequent, less intense and do not cause distress.
- ★ *Epidemiology*: Prevalence of OCD in children and adolescents is estimated as 0.5%. OCD may occur as early as 5 years of age, though the mean age of onset is 10 years. Childhood onset OCD has a ratio of 2 boys to every girl. The onset of OCD post-pubertal is more common in girls.
- ★ *Aetiology*: Increased incidence in 1st degree relatives and in monozygotic twins (80% compared to 40% in monozygotic twins); possible autoimmune process; reduced volume of caudate nucleus; hyperactive orbitofrontal circuits (SPET); 5HT/serotonin receptor dysfunction (clinical response to SSRI's in both children and adults), and dopaminergic dysfunction [high dose stimulants may increase OCD symptoms]
- ★ *Symptoms*: Contamination fears with ritualized washing and avoidance; repetitive doubting and checking (e.g. locks); repetitive counting, arranging, hoarding or touching.
- ★ *Comorbidity*: 70% have at least one co morbid disorder. OCD is commonly found to be comorbid with other psychiatric disorders, such as Tic disorders (17-40%), major depression (26%) specific developmental disabilities (24%), phobias and other anxiety disorders. There are also higher than expected rates of attention-deficit/hyperactivity disorder (ADHD), autism spectrum disorders, and tic disorders, including Tourette's syndrome, among children and adolescents with OCD. DSM IV allows a diagnosis of OCD in the presence of schizophrenia.
- ★ *Treatment*: Results from multiple randomised placebo-controlled trials of both medication and CBT interventions in children and adolescents with OCD have confirmed the most evidence for successful treatment of this disorder compared to any of the other anxiety disorders of childhood. A multi-site National Institute of Health funded investigation of Sertraline and CBT each alone, and then in combination for the treatment of childhood onset OCD-the Paediatric OCD Treatment Study (POTS)-revealed that the combination was superior to either treatment alone
- ★ Fluoxetine and Sertraline are licensed for the treatment of OCD in children and adolescents, and should be prescribed as per BNF guidelines. (NICE guidelines)

- ★ Fluoxetine is recommended for OCD in young people with significant co-morbid depression.
- ★ Clomipramine is not recommended as a first line treatment due to its greater potential risks compared to other SSRI agents, including cardiovascular risk of hypotension and arrhythmia, and seizure risk
- ★ CBT geared toward children of varying ages is based on the principle of developmentally appropriate exposure to the feared stimulus coupled with response prevention, leading to diminishing anxiety over time experienced upon exposure to feared situations
- ★ Most treatment guidelines for children and adolescents with mild to moderate OCD recommend a trial of CBT prior to initiating medication. However, there is evidence from POTS that optimal treatment includes the combination of both SSRI medication and CBT
- ★ Young people with mild functional impairment are initially offered self help guides. If self help hasn't worked, or for those with mild and moderate impairment, CBT [including exposure response prevention] should be offered. In the event of an inadequate response, or for those with severe impairment, supplement psychotherapy with SSRI should be considered.

PANDAS syndrome

- ★ Some children appear to develop obsessive-compulsive symptoms associated with beta haemolytic streptococcal infection and this presentation represents a minority of OCD cases in this population. This association has led to the studies of immune responses in OCD. Cases of infection triggered OCD have been termed paediatric autoimmune neuropsychiatric disorders associated with streptococcus (PANDAS) and believed to signify an autoimmune process such as that of Sydenham's chorea during rheumatic fever. It is associated with OCD and/or Tic Disorder.
- ★ Pre-pubertal onset is common.
- ★ 75% of children and young people with Sydenham's chorea also have OCD symptoms. Specific characteristics include episodic or 'sawtooth' course, choreiform movements. Typically, symptoms arise along with tics and this phenomenon may be related to obsessive-compulsive symptoms seen in Sydenham's chorea. It is very common for children with tic disorders and perhaps OCD to report an abrupt onset and experience exacerbation of symptoms with infection. Singer and co-workers reported that over 50% of clinic patients with tic disorders reported this finding and that 11% reported exacerbation of illness within 6 months of streptococcal infection
- ★ It is hypothesized that exposure to streptococcal bacteria activates the immune system leading to inflammation of the basal ganglia and resulting disruption of the cortical-striatal-thalamo-cortical function.
- ★ Scanning data suggest that increased basal ganglia volumes are associated with PANDAS; just as they are in OCD, tic disorders and Sydenham's chorea. MRI has documented a proportional relationship between the size of the basal ganglia and the severity of OCD symptoms.

Elimination Disorders

Enuresis

- ★ Enuresis is the repeated voiding of urine into a child's clothes or bed; the voiding may be involuntary or intentional. For the diagnosis to be made, a child must exhibit a developmental or chronological age of at least 5 years. Duration of disorder is at least 3 months.
- ★ Voluntary control of micturition does not begin until 15-18 months. Most children are reasonably dry by day at 18 months. By 2 years 50% are dry at night, by 3 years 75% are dry at night. By 5 years, only 1% of children have problems with daytime wetting. However, nocturnal enuresis can still affect 15-

22% boys and 7-15% girls at the age of 7 years. A small minority continues to have problems into adulthood.

- ★ Enuresis can occur as a psychiatric complication of stressful life events or emotional disturbance; it is not always accompanied by encopresis. It is twice as common in boys than girls. Types: Primary or secondary, nocturnal or diurnal or mixed.
- ★ Reported prevalence: 2-5% among school-aged children.
- ★ *Nocturnal enuresis*: Most important predictor of primary nocturnal enuresis is a family history of enuresis. 70% of children with nocturnal enuresis have a parent or sibling who was late in becoming dry. Secondary enuresis is predicted by delay in control over bedwetting, and experiencing a high rate of adverse life events. There is also a reduced sensitivity to vasopressin [antidiuretic effect] in the kidneys.
- ★ *Daytime enuresis* is likely to be related to structural abnormalities. Sexual abuse is associated with secondary wetting. Enuresis is diagnosed only if the behaviour is not caused by a medical condition. It is important to rule out organic factors, such as the presence of urinary tract infections, obstructions, genitourinary pathology, neurological conditions like spina bifida occulta; other organic disorders that can cause polyuria and enuresis, such as diabetes mellitus and diabetes insipidus.
- ★ Association of enuresis has been made with a high rate of stressful life events, UTI, constipation, and with children from low socioeconomic backgrounds, living in large families or overcrowded conditions.
- ★ Enuresis is often self-limited, and a child with enuresis may have a spontaneous remission without psychological sequelae. The significant emotional and social difficulties of these children usually include poor self-image, decreased self-esteem, social embarrassment and restriction, and intrafamilial conflict. Children with enuresis are at higher risk for ADHD compared with the general population. They are also more likely to have comorbid encopresis

Treatment

- ★ Rule out organic causes of urinary dysfunction –e.g.: UTI may lead to obstruction. The first step in any treatment plan is to review appropriate toilet training.
- ★ *Psychoeducation*: Child with nocturnal enuresis is not being lazy. Avoid punishments.
- ★ Record keeping is helpful in determining a baseline and following the child's progress and may itself be a reinforcer.
- ★ *Behavioural interventions* are the first line of treatment for bed-wetting. Bell and pad, or enuresis alarm. Classic conditioning with the bell and pad apparatus is generally the most effective treatment for nocturnal enuresis, with dryness resulting in more than 60 percent of cases. A star chart may be particularly helpful.
- ★ *Medications*: Imipramine is efficacious and has been approved for use in treating childhood enuresis, primarily on a short-term basis. Desmopressin, an antidiuretic compound that is available as an intranasal spray, has shown some initial success in reducing enuresis. Adverse effects that can occur with Desmopressin include headache, nasal congestion, epistaxis, and stomachache. One case of hyponatraemic convulsion. Reboxetine a norepinephrine reuptake inhibitor with a non cardiotoxic

side effect profile has recently been investigated as a safer alternative to Imipramine in the treatment of childhood enuresis. Oxybutynin is an anticholinergic, which increases bladder capacity though rarely employed in clinical practice.

Encopresis

- ★ Voluntary or involuntary soiling of normally formed stools in inappropriate places in a child of 4 years (mental age) or older, in the absence of a sufficient organic cause. Soiling occurs at least once a month for 6 months.
- ★ Males are found to be about six times more likely to have encopresis than females. Approximately 5% children will be soiling over 4 years of age.
- ★ Parents present more often to paediatrics than child psychiatric services. Although encopresis is considered a nonorganic disorder, more than three quarters of children with encopresis may show evidence of chronic constipation, leading to infrequent defecation, withholding of bowel movements, which then leads to impaction and eventual overflow soiling. However, it is important to exclude organic cause such as Hirschprung's disease, anorectal pathology, neurological problems; nutritional disorders and medication side effects.
- ★ Encopresis has been demonstrated to occur with significantly greater frequency among children with known sexual abuse compared with a normal sample of children. This can be seen as possible defense against further abuse, or retention due to pain.
- ★ It occurs with greater frequency among children with a variety of psychiatric disturbances compared with controls and some evidence indicates that encopresis in children is associated with measures of maternal hostility, and harsh and punitive parenting.
- ★ Encopresis, in some children, can be considered secondary, that is, emerging after a period of normal bowel habits. Family patterns such as an unhappy child in a family with clear ongoing difficulties [abuse, domestic violence], recent acute stress in the family [death, birth of sibling etc.], and over tolerant parents have been found in half the families studied in an audit [Silver,1996].

Treatment

- ★ Initial paediatric assessment, including imaging of the gut if indicated. First line of treatment is to evacuate any stool. Laxatives may be used to prevent further retention.
- ★ Education of the family and correction of misperceptions that a family may have about soiling must occur before treatment
- ★ Psychological treatment: Behavioural approach – operant training with rewards and positive reinforcement.
- ★ Family support/therapy. It is important to reduce the family tensions about the symptom and a nonpunitive atmosphere is established. Randomised control trials show that biofeedback adds nothing to laxative treatment.
- ★ In many cases, encopresis is self-limiting, and it rarely continues beyond middle adolescence. Most soiling stops by the age of 16 years.

Pica

- ★ This condition is characterised by persistent (>1month) eating of non-nutritive substances at a developmentally inappropriate age (>1 year) and it occurs at least twice a week.
- ★ It typically occurs during the 2nd and 3rd years of life and common substances are: dirt, stones, plastic, paper, hair, faeces, wood etc. It is common in people with developmental disability.
- ★ Consequences may include toxicity, infection, or GIT ulceration/obstruction.
- ★ Hypothesized causes include: hunger, malnutrition, nutritional deficiencies, psychosocial stressors and brain disorders such as lesions in the hypothalamus

8. Developmental disorders

Pervasive developmental disorders (PDD)

- ★ Pervasive developmental disorders include several that are characterized by impaired reciprocal social interactions, communication difficulties, aberrant language development, and restricted behavioural repertoire. They typically emerge in young children before the age of 3 years. Parents often become concerned about a child by 18 months if language development does not occur as expected. PDD is an umbrella term, which includes disorders such as childhood autism, Asperger's syndrome and Rett's syndrome.
- ★ PDD's frequently involve a triad of deficits in social skills, communication/language, and behaviour. The feature that all have in common is a difficulty with social behaviour.
- ★ DSM-IV categorised PDDs as follows: Autism, Asperger's syndrome, Rett's syndrome, Childhood disintegrative disorder and PDD-NOS. In DSM-5, Autism Spectrum Disorder is a new description that will now include autism, Asperger's, Childhood Disintegrative Disorder, and Pervasive Developmental Disorder (not otherwise specified) in a single category. ASD is characterized by 1) deficits in social communication and social interaction and 2) restricted repetitive behaviors, interests, and activities (RRBs). If no RRBs are seen, then a diagnosis of social communication disorder can be applied.

Autism

- ★ Childhood autism (previously called early infantile autism, Kanner's autism) is characterized by features from the following three symptom domains:
 - Qualitative impairment in social interaction
 - Impairment in communication [language delay is most common cause for initial referral]
 - Restricted repetitive and stereotyped patterns of behaviour or interests.
- ★ The failure of those with autism to understand others' situations or feelings is often referred to as a lack of theory of mind.
- ★ By definition, the onset of childhood autism is before the age of 3 years. In some cases, a diagnosis is not made until a child is much older (sometimes in adulthood).
- ★ Autism is four times more frequent in boys than in girls. There has been an apparent increase in prevalence rates over the years.

Aetiology and Pathogenesis

Genetic Factors

- ★ Family studies: Increased risk of autistic spectrum disorder in siblings of those with ASD, compared to general population. The recurrence rate in siblings of affected children is 2-8%. The risk of autistic disorder in a sibling of 2 autistic children is 25-30%.
- ★ Autistic spectrum disorders are subject to a significant degree of heritability (overall 90% heritability). The concordance rate of autistic disorder in the two largest twin studies was 36% vs. 0% in monozygotes vs. dizygotes in one study and about 96% vs. 27% in another.
- ★ It now appears that multiple genes are involved in the development of autism and linkage analyses have demonstrated that regions of chromosomes 2, 4, 7, 13, 15, and 19 are likely to contribute to the genetic basis of autism. Likely locations for autism-related genes were also found on chromosomes 16 and 17, although the strength of the correlation was somewhat weaker. Some analyses have implicated genes such as NRXN1 and NLGN3 (Short Oxford Textbook of psychiatry, 3rd Edn).
- ★ Fragile X syndrome, a genetic disorder in which a portion of the X chromosome fractures, appears to be associated with autistic disorder. Approximately 1-4 percent of children with autistic disorder also have fragile X syndrome. Tuberous sclerosis, a genetic disorder characterized by multiple benign tumours, with Autosomal dominant transmission is found with greater frequency among children with autistic disorder. Up to 2 percent of children with autistic disorder may also have tuberous sclerosis.
- ★ Autistic disorder is also associated with neurological conditions, notably congenital Rubella and phenylketonuria

Biological Factors

- ★ About 10% of children with autism (1 in 10 cases) have a medical condition of some type (Shorter Oxford Textbook of Psychiatry, 6th edn). The high rate of intellectual disability among children with autistic disorder and the higher-than-expected rates of seizure disorders further support the biological basis for autistic disorder.
- ★ Approximately 80% of people with childhood autism have learning disability. About one third of these children have mild to moderate intellectual disability. In contrast, about 80% of people with Autism Spectrum Disorders more generally have intellectual abilities within the normal range.
- ★ MRI findings include larger brain volumes and early acceleration in brain growth; increase in the size of the lateral and 4th ventricles; frontal lobe and cerebellar abnormalities (hypoplasia of cerebellar vermal lobules VI and VII). The greatest average percentage increase in brain size occurred in the occipital lobe, parietal lobe, and temporal lobe. No differences were found in the frontal lobes. The increased volume can arise from three different possible mechanisms: increased neurogenesis, decreased neuronal death, and increased production of nonneuronal brain tissue, such as glial cells or blood vessels. Brain enlargement has been suggested as a possible biological marker for autistic disorder.
- ★ Cerebellar pathology: abnormal purkinje cells in cerebellar vermis; abnormal limbic architecture. An autopsy study revealed fewer Purkinje's cells.

- ★ A number of studies have demonstrated that about one third of patients with autistic disorder have high plasma serotonin concentrations. In some autistic children, a high concentration of homovanillic acid in cerebrospinal fluid (CSF) is associated with increased withdrawal and stereotypes.

Prenatal and postnatal factors:

- ★ There are various associations between autistic spectrum disorder and congenital rubella infection during pregnancy. Most associations are with first trimester exposure, suggesting that an impact on early brain development is of particular significance. Obstetric complications are currently not thought to represent a causative factor
- ★ Child rearing practices do not account for autism.
- ★ MMR vaccination was hypothesised to be related to a regressive form of autism. These links arose due to the findings of a single methodologically flawed study. Other studies that aimed to replicate these findings were unable to do so. (Ref: The Handbook of Child and adolescent Psychiatry-pg 132)

Course and Prognosis

- ★ Autism is a neurodevelopmental disorder that persists lifelong. Predictors of good prognosis include communicative speech by the age of 6 years and higher IQ (>50) and having a skill that is consistent with a secure employment. A significant proportion of people with autism (at least 60%) will be unable to lead an independent life, and some will require long term residential care (Shorter Oxford Textbook of Psychiatry, 6th edn).
- ★ Findings of long term follow up studies into adult life- 12% very good outcome, 10% as good, 19% as fair, 46% as poor, 12% as very poor. Communication, reading and spelling were impaired in most. Stereotypies and restricted interests remained for most. (Ref: The Handbook of Child and adolescent Psychiatry-pg 140)

Treatment

- ★ There is no cure or treatment for autism. Early and effective management is desirable for a better outcome, and involves a comprehensive assessment of the child and family's needs. There are many management models, aimed at helping the family, providing vocational training and support to the person, addressing health requirements etc. Psychotherapy with the individual may be in the form of specific CBT techniques for those with verbal skills, and behaviour management programmes. However, young people with autism can find CBT difficult due to their tendency to sometimes have literal understanding and rigid thinking.
- ★ Two specific intervention programmes are
 - Applied Behavioural Analysis program: an intense program [40 hours a week for 3 years] based on operant conditioning, imitation and reinforcement.
 - TEACCH – Treatment and Education for Autistic and related Communication Handicapped Children program is based on the belief that children are motivated to learn language. Successful in reducing self injurious behaviour, and enhancing life skills.

- ★ Pharmacotherapies are commonly used in individuals with autistic spectrum disorder (ASD) as adjuncts to psychological interventions. SSRIs have become the most widely prescribed medications to treat restricted repetitive behaviours in paediatric ASD populations. However the evidence supporting the effectiveness of SSRIs in ameliorating these symptoms remains limited. When using SSRIs to treat repetitive behaviour in ASD patients, it has been found that lower doses are required than the therapeutic antidepressant dose. Second generation antipsychotics are the first line pharmacological treatment for children and adolescents with ASD and associated irritability. The only licensed medication in UK is risperidone. It is indicated for the treatment of autism with aggressive behaviour in children and adolescents. Weight gain, somnolence and hyperglycemia require monitoring. Melatonin has been shown in several small studies to be beneficial in children with ASD with reductions in sleep latency as well as efficiency.

Asperger's syndrome

- ★ Individuals with Asperger's syndrome have same type of qualitative abnormalities in reciprocal social interaction that typify autism with a restricted, stereotyped and repetitive, repertoire of behaviour/interests. IQ and language are usually within normal limits.
- ★ Unlike in childhood autism/autistic disorder, there were in Asperger's disorder no significant delays occur in relation to language development, cognitive development, and/or age-appropriate self-help skills.
- ★ Mild motor clumsiness and a family history of autism may be present, as is the case for other autism spectrum disorders.
- ★ Epidemiology: Global prevalence rate is uncertain. Estimated to be about a third as common as autism with a prevalence of about 6 in 10,000 (Shorter Oxford Textbook of Psychiatry, 6th edn).
- ★ The factors associated with a good prognosis are a normal IQ and high-level social skills. Psychiatric comorbidity is high with depression most common. BPAD and schizophrenia are more common than in the general population.

Rett's syndrome

- ★ Rett's syndrome is a rare X linked dominant disorder of arrested neurodevelopment associated with mutations in the MeCP2 gene. It almost exclusively affects females.
- ★ Head circumference normal at birth and developmental milestones are unremarkable in early life. Between 6 and 18 months, head growth begins to decelerate and produces microcephaly. Other signs often include the loss of purposeful hand movements, which are replaced by stereotypic motions, such as hand wringing; the loss of previously acquired speech; psychomotor retardation; and ataxia.
- ★ Other stereotypical hand movements may occur, such as licking or biting the fingers and tapping or slapping. Language skills are lost, and both receptive and expressive communicative and social skills seem to plateau at developmental levels between 6 months and 1 year. Poor muscle coordination and an apraxia gait with an unsteady and stiff quality develop.
- ★ Associated features include seizures in up to 75 percent of affected children and disorganized EEGs with some epileptiform discharges in almost all young children with Rett's disorder, even in the absence of clinical seizures.

- ★ An additional associated feature is irregular respiration, with episodes of hyperventilation, apnoea, and breath holding. The disorganized breathing occurs in most patients while they are awake; during sleep, the breathing usually normalizes. Many patients with Rett's disorder also have scoliosis and spasticity.
- ★ Developmental arrest plateaus and many may reach adulthood, although disability is great and prognosis poor.

Childhood disintegrative disorder (CDD)

- ★ Childhood disintegrative disorder (CDD) also called Heller's disease and disintegrative psychosis is characterized by marked regression in several areas of functioning after at least 2 years of apparently normal development.
- ★ There is normal development for 2-3 years, followed by a loss of acquired motor, language, and social skills between ages 3 and 4 years. Stereotypies and compulsions are common
- ★ To receive the diagnosis, a child must exhibit loss of skills in two of the following areas: language, social or adaptive behaviour; bowel or bladder control; play; and motor skills. Abnormalities must be present in at least two of the following categories: reciprocal social interaction, communication skills, and stereotyped or restricted behaviour. The main neurological associated feature is seizure disorder
- ★ Male predominance is noted. Cause is unknown and prognosis is poor. Most patients are left with at least moderate mental retardation

9. Learning and communication disorders

- ★ Learning disorders are problems that make educational achievement difficult. Specific learning difficulties can be underachievement in reading, written expression, or mathematics in comparison with the overall intellectual ability of the child.
- ★ The most recent revised version of the DSM-IV (DSM-IV-TR) includes four diagnostic categories of learning disorders: reading disorder, mathematics disorder, disorder of written expression, and learning disorder not otherwise specified
- ★ Genetic predisposition, perinatal injury, and neurological and other medical conditions can contribute to the development of learning disorders, but many children and adolescents with learning disorders have no specific risk factors
- ★ Learning disorders affect at least 5% of school-age children. They are associated with higher than average risk of a variety of co morbid disorders, including attention-deficit/hyperactivity disorder (ADHD), communication disorders, conduct disorders, and depressive disorders.
- ★ DSM-IV groups together a number of learning disorders that share the following:
 - Performance significantly below that expected for IQ or age
 - A discrete developmental disability in the absence of learning disability
 - Commonly present as emotional or behavioural problems
 - 50% have a comorbid psychiatric disorder and many have other LLDs
 - Most show strong evidence of heritability
- ★ Communication disorders are among the most common disorders in childhood. They include language disorders such as expressive and mixed receptive-expressive language disorder, and speech disorders such as phonological disorder and stuttering.

Reading disorder (dyslexia)

- ★ Reading disorders are present in approximately 75% of children and adolescents with learning disorders. Reading disorder is characterized by an impaired ability to recognize words, slow and inaccurate reading, and poor comprehension. Children who have reading disorder can usually be identified by the age of 7 years (second grade).
- ★ Reading comprehension skill, reading word recognition, oral reading skill, and performance of task requiring reading may all be affected. Children with reading disorders make many errors in their oral reading. The errors are characterized by omissions, additions, and distortions of words. Such children have difficulty in distinguishing between printed letter characters and sizes, especially those that differ only in spatial orientation and length of line. The child's reading speed is slow, often with minimal comprehension. Difficulty with reading, in most cases involving a deficit in phonological-processing skills.
- ★ 4% of school-age children are affected. Male predominance [4:1]. There is often a family history of dyslexia. Approximately 20% of children with reading disorder have comorbid CD or ADHD (Oxford Handbook of Psychiatry, 3rd Edn.). Conversely, it is estimated that between 15 and 30 percent of children diagnosed with ADHD have a learning disorder.

- ★ Tools used to measure reading ability are WORD (Weschler objective reading dimension)- a single word rreading test commonly used in Bristish schools, TOWRE (Test of word reading efficiency)- measures word reading rate & accuracy, WISC (Wescler intelligence sacle for children)- measures overall cognitive ability of the child
- ★ Management includes 1:1 remedial teaching and parent involvement improves long-term outcome.

Disorder of written expression

- ★ Disorder of written expression is characterized by writing skills that are significantly below the expected level for a child's age and intellectual capacity. These difficulties impair the child's academic performance and writing in everyday life.
- ★ Common features of the disorder are spelling errors, grammatical errors, punctuation errors, poor paragraph organization, and poor handwriting. Often coexists with dyslexia and manifests as difficulties with spelling, syntax, grammar, and composition.
- ★ Occurs in 2-8% of school-age children with a 3:1 male predominance.
- ★ ADHD occurs with greater frequency in children with writing disorders than in the general population.
- ★ Remedial treatment for writing disorder includes direct practice in spelling and sentence writing as well as a review of grammatical rules. Intensive and continuous administration of individually tailored, one-on-one expressive and creative writing therapy appears to effect favourable outcome.

Mathematics disorder

- ★ Epidemiological studies have indicated that up to 6 percent of school age children have some difficulty with mathematics. It is reported more commonly in females.
- ★ Often associated with visuo-spatial deficits and attributed to right parietal dysfunction.
- ★ Common features of mathematics disorder include difficulty with various components of mathematics, such as learning number names, remembering the signs for addition and subtraction, learning multiplication tables, translating word problems into computations, and doing calculations at the expected pace. A Child with mathematics disorder generally has significant problems with concepts, such as counting and adding even one-digit numbers, compared with classmates of the same age
- ★ Currently, the most effective treatments for mathematics disorder combine teaching mathematics concepts with continuous practice in solving math problems. Flash cards, workbooks, and computer games can be a viable part of this treatment

Learning Disorder Not Otherwise Specified:

- ★ Learning disorder not otherwise specified- It does not meet the criteria for any specific learning disorder, but causes impairment and reflects learning abilities below those expected for a person's intelligence, education, and age. An example of a disability that could be placed in this category is a spelling skills deficit.
- ★ The ICD-10 classifies specific developmental disorders of scholastic skills learning disorders that include specific reading disorder; specific spelling disorder; specific disorder of arithmetic skills;

mixed disorder of scholastic skills; other developmental disorders of scholastic skills; and developmental disorder of scholastic skills, unspecified

Expressive language disorders

- ★ Expressive language disorder is diagnosed when a child demonstrates a selective deficit in expressive language development relative to receptive language skills and nonverbal intelligence.
- ★ A child with expressive language disorder is likely to function below the expected levels of acquired vocabulary, correct tense usage, complex sentence constructions, and word recall. A child with an expressive language disorder exhibits the following features: limited vocabulary, simple grammar, and variable articulation. Language understanding (decoding) skills remain relatively intact.
- ★ Prevalence: a rate of between 3-5% of children has been suggested (Shorter Oxford Textbook of Psychiatry, 6th Edn). The disorder is two to three times more common in boys than in girls and is most prevalent among children whose relatives have a family history of phonological disorder or other communication disorders.
- ★ The most common co morbid disorders were attention-deficithyperactivity disorder (ADHD), anxiety disorders, oppositional defiant disorder, and conduct disorder.
- ★ Management includes special education, speech therapy and treating any comorbidity, if applicable.

Mixed receptive-expressive language disorder

- ★ The essential clinical feature of the disorder is significant impairment in both language comprehension and language expression. In the mixed disorder, the expressive impairments are similar to those of expressive language disorder, but can be more severe.
- ★ Children with mixed receptive-expressive language disorder show markedly delayed and below-normal ability to comprehend (decode) verbal or sign language, although they have age-appropriate nonverbal intellectual capacity. In most cases of receptive dysfunction, verbal or sign expression (encoding) of language is also impaired.
- ★ Mixed receptive-expressive language disorder tends to persist in approximately 3 percent of school-age children and is believed to be at least twice as prevalent in boys as in girls.

Phonoogical disorder

- ★ Children with phonological disorder are delayed in, or incapable of, producing speech sounds that are expected for their age, intelligence. Their inability to produce clear speech is not due to physical problems as may be seen in dysarthria.
- ★ Phonological disorder can be recognized in early childhood. In severe cases, the disorder is first recognized at about 3 years of age.
- ★ Omissions, distortions, and substitutions also occur normally in the speech of young children learning to talk. But, whereas young, normally speaking children soon replace these misarticulations, children with phonological disorder do not.
- ★ Children with phonological disorder cannot articulate certain phonemes correctly and may distort, substitute, or even omit the affected phonemes. With omissions, the phonemes are absent entirely for example ca for car, or whaa? for what's that? With substitutions, difficult phonemes are replaced with

incorrect ones for example, wabbit for rabbit. With distortions, the correct phoneme is approximated but is articulated incorrectly.

Stuttering

- ★ Stuttering is a condition in which the normal flow of speech is disrupted by involuntary speech motor events. Stuttering can include a variety of specific disruptions of fluency, including sound or syllable repetitions, sound prolongations, dysrhythmic phonation, and complete blocking or unusual pauses between sounds and syllables of words. Usually struggle with initial syllables.
- ★ Male predominance of 3:1 is noted. The prevalence of stuttering is about 1% in the general population. Stuttering tends to be most common in young children and has often resolved spontaneously by the time the child is older. The typical age of onset is 2 to 7 years of age with a peak at age 5 years. Of all children who stutter, mostly those with mild cases, 50 to 80 percent recover spontaneously. Most children grow out of it but stuttering affects 1-2% of adults.
- ★ Preschoolers and school-age children who stutter exhibit an increased incidence of social anxiety, school refusal, and other anxiety symptoms. When stuttering persists into adolescence, social isolation occurs at higher rates than in the general adolescent population.
- ★ Stuttering is also associated with a variety of abnormal motor movements, upper body tics, and facial grimaces. Other disorders that coexist with stuttering include phonological disorder, expressive language disorder, mixed receptive-expressive language disorder, and ADHD.
- ★ However, stuttering is not normally associated with a psychiatric disorder (Shorter Oxford textbook of Psychiatry, 6th edn.). Majority of cases are developmental but occasionally there are acquired cases (e.g. head injury). Aetiology: genetic; incomplete cerebral dominance; hyperdopaminergic state.
- ★ Management: speech therapy.

10. Tic disorders

Tourette syndrome

- ★ Characterised by multiple motor and one or more vocal tics, present for at least a year, causing distress and impaired function.
- ★ The tics occur many times a day (usually in bouts) nearly every day or intermittently throughout a period of more than 1 year, and during this period there was never a tic-free period of more than 3 consecutive months.
- ★ Facial tics: eye blinks or head jerks are often the initial symptoms, but tics involving the neck, shoulders, and upper extremities are also common.
- ★ Vocal tics begin 1-2 yrs after the onset of motor symptoms and range from meaningless sounds to clear words. The vocal tics are crucial for diagnosis and may be barks, hisses, hoots and not necessarily obscene blasphemous utterances (Coprolalia), echolalia, or palilalia (repeating one's own speech)
- ★ Approximately 10% exhibit echolalia or echopraxia. About 1/3 exhibit coprolalia (but most of these are not children) (Shorter Oxford Textbook of Psychiatry, 6th edn).
- ★ Typically, tics vary over time with more complex tics emerging after some years.

- ★ Wax and wane in frequency with periods of remission. The periods of exacerbations are often related to physical and emotional stress. These are known to be exacerbated by external factors such as stress, anxiety and fatigue
- ★ The onset is before age 18 years. Suggested prevalence in children of 1%. Monozygotic twin concordance 50%, dizygotic 10%. Mean age of onset is 7 years. It is 3-4 times more likely in males than females (Shorter Oxford Textbook of Psychiatry, 6th edn).

Aetiology

- ★ *Genetic factors*: Autosomal dominant and polygenic common (with increase in Monozygotic twins and first degree relatives). Family history very common. Overlap with genetic liability for OCD.
- ★ *Neurotransmitter abnormalities* thought to underlie tic disorders. Functional excess of dopamine system is a popular speculation as antipsychotics reduce symptoms. The effectiveness of Clonidine suggests the role of Noradrenaline
- ★ *Neuroimaging* studies implicate circuits involving the basal ganglia, premotor and motor cortex.
- ★ *Post-infectious autoimmune mechanisms*: It is well established that group A beta haemolytic streptococci (GABHS) can trigger immune-mediated disease in genetically predisposed individuals. Speculation concerning a post-infectious (or at least a post-rheumatic fever) etiology for tic disorder symptoms dates from the late 1800s. An increased proportion of GABHS infections (odds ratio = 3.1) is found in the preceding 3 months in children with newly diagnosed TS compared to well matched controls. Larger odds ratio (OR = 12.1 for TS) is demonstrated when subjects were required to have multiple GABHS infections in the previous year, suggesting a dose-dependent effect of exposure. There was also a statistically significant positive association between preceding streptococcal infection and a new diagnosis of OCD and tic disorders.
- ★ **Comorbidity**: OCD [1/3rd to 2/3rd of children and adolescents exhibit OCD symptoms) and ADHD [upto 50%] common; depression, anxiety, impulse and conduct disorders and other behavioural problems. These are more prevalent than would be expected by chance association alone. 1/3rd of adults with Tourettes have persistent OCD into adulthood.

Management:

- ★ Psychoeducation for the individual, family and the people they interact with, especially schools is crucial. Both tics and OCD symptoms may improve with CBT, if successful exposure-response prevention can be achieved. Techniques include a) Relaxation b) Exposure-response prevention c) massed practice (forced repetition of the tic) d) Habit reversal (movements incompatible with the tic) (Ref: The Oxford handbook of child and adolescent psychiatry; pg 198)
- ★ Medications can be used if tics are disabling and non-responsive to other therapies. Medications used to treat Tourette's disorder include Haloperidol, Pimozide, Sulpiride, Risperidone and Clonidine at low doses. All these drugs have been shown to be more effective than placebo in a number of small-randomised studies.
- ★ Clonidine (adrenergic alpha 2 agonists) should be tried first. It helps to reduce the severity and frequency of tics. Main side effects are sedation and postural hypotension. Guanfacine also an

adrenergic alpha 2 agonists has been shown to lead to a 30% reduction in tic rating scales. These drugs are more acceptable in terms of adverse effects.

- ★ Antipsychotics like Risperidone, Haloperidol and sulpiride are effective in alleviating tics but their side effects may outweigh benefits and should not be tried as first line drugs.

Transient Tic Disorder

- ★ Transient tic disorder is characterised by single or multiple motor and/or vocal tics (sudden, rapid, recurrent, nonrhythmic, stereotyped motor movements or vocalizations) affecting usually face, nose, throat are involved (grimaces, blinks, frowns, grunts, throat clearing, sniffs). The tics occur many times a day, nearly every day for at least 4 weeks, but for no longer than 12 consecutive months
- ★ The onset is before age 18 years. The disturbance is not due to the direct physiological effects of a substance (stimulants) or a general medical condition (Huntington's disease or post viral encephalitis).
- ★ Criteria should have never been met for Tourette's Disorder or Chronic Motor or Vocal Tic Disorder.
- ★ Note: Transient tics occur in 5-20% of children. 80% of children with tics will have outgrown them by adult life.

Chronic Tic disorder:

- ★ Persists for more than one year and often shows relapses and remissions throughout childhood. May be one or several tics are simultaneously present. Motor or vocal tics but not both (so it is a form of partial Tourette syndrome). Over time, one form fades to be replaced by another. It is a rarer form of tic disorder.

11. Miscellaneous topics

Substance abuse in the early life

- ★ *Alcohol* is the most commonly abused substance by adolescents. ~ 30% of 13- year olds admitting to the use of alcohol at least once a week. Binge drinking pattern is more common than other patterns of abuse among the adolescents
- ★ *Tobacco* use (cigarette smoking) in young people is decreasing in the West due to legislative measures and changing public perception
- ★ *Cannabis*: ~ 5% of 11-16 year olds in the UK had used cannabis at least once previously, with rates increasing significantly in the late teens.
- ★ *Heroin* use in under 18's has increased in the past decade in many countries including the UK. At least 15% of users will switch to IV use within 1 year, with its attendant additional risks.
- ★ 2.4% of the 12-17 year old age group used *cocaine* at some point in their life. (US national survey) whereas 0.5% of 11-15 year olds in the UK admitted to using cocaine.
- ★ *Party drugs* like MDMA (ecstasy), GHB, ketamine, rohypnol) do not account for a large proportion of child and adolescent substance problems, but are being experimented with at increasingly young ages.
- ★ There is a downward trend of life-time *methamphetamine* use (speed) from 1999 to 2005 (4.7% rate reduced to 2.5% lifetime rate now)

- ★ In terms of *steroid* use, in the US, 4 and 12% of male adolescents have used anabolic steroids to improve performance at sport or alter body size. No UK data is available, but the rates are often reported to be less than the US.
- ★ Use of *LSD/hallucinogens* is down since a peak in the 1990s and now used only by a small minority.

(Ref: The Oxford handbook of Child and Adolescent Psychiatry: 382-383)

Rating scales in Children and Adolescents

Scale	Properties
Connor's questionnaire	It is widely used to obtain information from parents and schoolteachers on ADHD symptoms. There are parent and teacher versions.
Autism diagnostic interview- Revised (ADI-R)	It is a structured interview mainly used in research to obtain a thorough assessment of individuals suspected of having autism. It is useful for formal diagnosis as well as treatment and educational planning. It is composed of 93 items focusing on 3 areas; language & communication, reciprocal social interactions, restricted & stereotyped interests and behaviours.
Wechsler intelligence scale for children (WISC)	It is useful to assess intelligence for children aged less than sixteen. It has a verbal, performance subscales that yield a verbal, performance and full-scale IQ. WAIS is for adults.
Strength and difficulties questionnaire (SDQ)	It is a generic rating scale, helpful to obtain information on a variety of psychiatric symptoms like emotional, behavioural and prosocial problems. It is validated in a number of different populations of children aged between 4-16. It is quick to complete.
The Child Behavior Checklist (CBCL)	The Child Behavior Checklist (CBCL) assesses a broad range of symptoms that relate to academic and social competence of children aged 4 through 16. Mainly used in children with oppositional defiant disorder and conduct disorder. CBCL is completed by the parents or others (e.g., teachers) who know a child well. The checklist is composed of 113 items in a Likert fashion. This scale can be helpful in the evaluation of a child with multiple behavioural problems, especially if the child presents with different symptoms in different settings such as at school and home. There is a parent and a teacher version so that the reports of these two observers may be compared.
The Child & Adolescent Functional Assessment Scale	It is a rating scale to assess the degree of impairment in functioning due to emotional, behavioural, or psychiatric problems. It is completed by a clinical staff and takes about 10 minutes. It measures aggression and conduct problems especially in age between 7 to 17 years.
Children's Depression Inventory	It is a scale used to measure severity of depressive symptoms between 7 to 17 years.

Beck Depression Inventory	It could be used for screening of children with depression only after the age of 14.
CY-BOCS	It is the paediatric version of Y-BOCS (Yale Brown Obsessive-Compulsive Scale) useful to measure symptoms of childhood OCD

In-patient psychiatric care

(excerpt from Blanz & Schmidt 2000)

- ★ In general, psychiatric hospitalization is not a preferred option for the assessment and treatment of most childhood-onset and adolescent psychiatric disorders, as this is seen as disruptive to the normal developmental context provided by a child's home and school environments.
- ★ Dalton et al. (1989) cited the following factors to argue against hospitalisation:
 - Disruption of the child's family and community relationships
 - Expenses incurred by the family and health services
 - Reinforcement of parental denial or guilt
 - Sibling's confused and distorted perception of the issues treated
 - Removal of the child from the continuum of education
 - Predictable noxious stigmata and labeling
 - potential for unresolved transference and institutionalisation
- ★ Hersov (1994) proposed a set of admission criteria that specify the need for hospitalisation (numbers 1–3) and the need to remove a child from its home (numbers 4 and 5):
 1. For diagnostic work that cannot be obtained on an outpatient basis
 2. A severe psychiatric disorder with need for treatment by a multiprofessional team in a safe setting
 3. Impaired physical status of the child that requires skilled medical and nursing care
 4. Adverse environmental circumstances that preclude the child's improvement within the home or severely distorted family interaction that leads to progressive interference with the child's progress
 5. Gross overprotection by parents after a trauma/injury that precludes recovery.

DISCLAIMER: This material is developed from various revision notes assembled while preparing for MRCPsych exams. The content is periodically updated with excerpts from various published sources including peer-reviewed journals, websites, patient information leaflets and books. These sources are cited and acknowledged wherever possible; due to the structure of this material, acknowledgements have not been possible for every passage/fact that is common knowledge in psychiatry. We do not check the accuracy of drug-related information using external sources; no part of these notes should be used as prescribing information

Notes prepared using excerpts from:

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